Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 8 ~ BookStream #002.02

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chapter eight is called active inference and continuous time begins with that timeless quote
everything flows nothing stands still so what would you say about chapter eight
all right so this chapter uh probably is my most favorite chapter in the book uh because of my
uh own personal interests in um i don't know the process materials and so on but
uh yeah so chapter seven um acts as a really good uh starting point for anyone who wants to uh
develop the discrete time situations uh to model discrete time situations uh inactive within active
inference framework but in chapter eight uh we kind of get into get to uh model uh a bit more
more interesting or let's say more involving situations uh and they're not necessarily
uh kind of toy examples we saw at least at the beginning of chapter seven uh so
obviously as the title suggests uh this chapter deals with the continuous time situations so
uh in the in that case uh we need to maybe at this point um refresh our memory about what continuous
time situation involves uh by reading the relevant parts uh re reading or reviewing relevant parts of
chapter uh four so uh yeah in chapter four we saw that the generative model for continuous time situation
uh derives uh derives uh derives from uh the uh ito's stochastic calculus uh in terms of um putting
the whole process um into two elements uh two stochastic um two elements of stochastic equations uh one of which
uh is the actual uh the condition of actual uh states um or the behavior of the actual states and the other one
uh is the randomness that we need to account for in each real-time continuous time situations so that's
what we get here uh in equation eight eight point one and then uh building up from that equation uh we
uh it generalizes that equation to involve uh um i mean the fun the functionals of g and f instead of just
the um the single valued functions of g and f so uh then we get to
uh put that into uh the situation that can be used for describing the behavior of dynamical systems
which is a very well-known uh situation to use these kinds of uh stochastic equations uh and uh
it's widely studied how those uh those kinds of dynamics can be characterized uh especially in recent
bayesian mechanics paper by uh dalton saktevadevel and others uh so and then it gets to some more
specific examples such as uh latke-volterra dynamics and uh synchronicity and so on uh in order to show
how uh these kinds of uh dynamics uh can be uh elaborated upon uh and uh can be generalized
uh uh to uh and and enables them to characterize more complex uh situations so uh yeah that's a really
short and brief overview of the whole uh chapter maybe uh we can talk about a bit more details as we go
through great well said well i'm sure for another day the philosophical implications of eight seven and
eight and high road and low road and all these other parts of the textbook great topics um i i agree i
would see chapter eight as demonstrating continuity with some classical continuous time modeling motifs
from a few different areas of dynamical system science which is applied in like many many many fields but
these are some classic examples so figure 8.1 goes a little bit more into depth or at least into more
formalism detail about exactly what we saw in chapter five with the spinal reflex arc with the proprioceptive

data coming in and then a differential being calculated with the set point which reflects a descending prediction prediction from a decision making layer and that can be viewed as this kind of mechanics that plays out in a phase space in continuous time like a spring moving around with with someone making a certain path with an attractor and a spring being dragged around something in that area box 8.1 goes into a very fascinating topic do you want to describe it uh well uh it's uh maybe uh one of the most um thought-provoking uh

section pages of the whole book and if i remember correctly uh in all of the cohorts this particular box

uh uh i mean gives always gives rise uh to lots of questions uh because of some of the interesting and uh at least

initially counterintuitive uh claims uh here uh but uh i don't want to spoil it so um the uh but uh as a kind of spoiler alert uh it kind of uh gets to uh really interesting but uh alas very brief discussion about

the comparing uh these terms precision attention and sensory attenuation and uh the relation and uh similarities and difference between these two these uh three terms and how uh each understanding each of them is essential to understanding uh the other ones uh but as i said uh it's a really interesting topic

uh which uh gives rise to lots of uh discussions uh and uh i believe it's one of those uh topics that

uh that's worth looking a bit more uh looking into uh in some other literature as well great well said what a cliffhanger next they go to a classic model family called laca volterra these dynamics inherit from characterizations of predator prey dynamics in ecology so it's kind of a classical ecology model shown in figure 8.2 on the top it's actually the ecosystem model plants herbivores and carnivores which follow different kinds of oscillatory trends in continuous time and so that also has enabled it to be applied for other so-called winnerless competitions and that relates to topics like neural darwinism and also neural dynamics where things have kind of oscillatory relationships with each other which are being modeled as a continuous time underlying process with a lot of measurement noise and discretization through space and time those are the kinds of algorithms that spm explores more and there's laca volterra and a lot of other dynamical systems theory in spm so active inference kind of adds action and more to what was laid out

from a pure dynamical systems theory in spm here it really is just showing the ecology example and how you can project if you have three different species you can think about that motion in a cube or tetrahedron and then you could project onto kind of like uh looking at a lower dimensional manifold relating just two of the three

species and that evinces this kind of oscillatory but also moving behavior that gets connected in figure 8.3 to neurobiology what would you say about this

uh okay so here in figure 8.3 uh we see some applications of um a lot of alterodynamics uh so uh the uh left column uh here uh show uh represents um what happens uh in uh i mean uh in eye blinking eye blink uh conditioning uh so uh uh of course uh here we uh need to account for uh uh i mean the expected states

uh uh uh of uh of uh of the of the of the sequence of the sequences of events that uh happens in in uh the eye

blinking so uh the upper uh upper left figure shows the expectations uh in terms of time and then uh the parallel right hand side equation uh sorry right hand uh side uh figures uh shows the uh

the lot covalterra system uh that that is applied uh in the handwriting situation so as we can see

although uh the uh i mean mathematical technology is the same or at least the modeling technology is the same uh the outcome of each situation uh varies drastically in uh two distinct

uh uh two distinct uh neurobiological behavior uh not neurobiological but biological behavior uh so

yeah we can see how uh the same modeling framework can give rise to uh different uh outcomes uh based on uh what parameters uh is uh needs to be optimized what parameters

uh are selected for the modeling and so on uh so i believe it's a quite uh interesting uh example to compare

uh handwriting and the blinking uh together and uh how uh those can be compared to each other using the lot

curve lot cobaltator dynamics great thank you box 8.2 gives a variant on the learning here presented with

the formalism for continuous models kind of a technical aside section 8.4 is about generalized synchrony

so figure 8.4 is going to visualize one of the classic dynamical systems which is the lorens attractor

so what would you say about this figure okay so this section is uh truly interesting because uh when

one thinks of active inference um probably the first situations that comes to mind is uh the situations in

which we have uh quite well defined uh probability distributions for different parameters but as we can see

here uh in section 8.4 uh actually some of the formalism of active inference uh can be successfully used uh to characterize

even chaotic systems and pretty and in particular the way in which two chaotic systems can be synchronized with

each other so uh this is a classic example of a chaotic uh lorentz system uh uh and um it it derives

from it draws upon uh from uh from um some of professor person's earlier work on birdsong uh synchrony uh and as a side note any literature before 2016 is considered earlier

history in active inference literature because it evolves uh quite rapidly uh so uh yeah

uh this kind of synchrony between uh two chaotic systems uh can be interpreted as providing evidence or even uh

let's say uh a way to model uh a kind of primitive theory of mind uh in in the sense that how exactly can we

uh understand uh or can two two agents can uh trace each other's trajectories uh without

uh any i mean engaging in any direct exchange of uh observations uh between their uh internal and external states so uh yeah that's a really good uh example and i believe one of the most interesting

examples of how active inference can even uh account for these kinds of uh behavior so uh and the rest of the section goes into the details of how uh this kind of synchrony between um

um multi multi-scale uh birds multi-scale um lorentz systems can happen um and how can we uh formulate it mathematically in terms of continuous time active inference

awesome awesome and there's been more recent work on markov blankets and stochastic chaos but the bird example is a classic

8.5 goes into hybrid discrete and continuous models so this could be kind of like a in-between chapter of

seven and eight but now that we've been introduced to the pure form of discrete and the pure form of continuous models here shown that that composability extends to so-called hybrid models where here the lower level visually is using the continuous time

formalism and the higher level is describing the little line added here the discrete time formalism and this was the similar structure

the similar structure described by the authors of the paper active inference does not contradict folk psychology

where they described this lower level as motor active inference which was closely allied with the spinal arc reflex

shown above and then this higher level they called decision active inference because in that case it was referring to a discrete decision

and so they used that kind of basic motif of continuous activity or continuous time modeling

at the more peripheral aspects of a cognitive entity and like ali said more discretization and hybridization as well

at higher levels of the cognitive modeling and that type of an architecture here instead of a

!

a new set point or fixed point being specified from the top down muscle command about a new location for a muscle

followed by movement towards it this is a muscular activity that is realizing that but but not in the elbow coming away from the hot stove

this is about the eye circading to an epistemic foraging location specified by top down hierarchical systems

8.3 describes a little technical aside on mixture of gaussian gaussian mixture models kind of a

technical modeling note and 8.6 closes it says it's a huge topic and much has been left out and so they list in table 8.1 key advances in continuous time models and those areas are

synthetic birdsong

oculomotor delays

conditioned reflexes

smooth pursuit eye movements

saccades

action observation

hybrid models

and self-organization

and that's chapter 8.

what else would you say and also what would you kind of lead someone to in the philosophical implications of 8.1 key

because it sounds kind of cool
uh okay so
uh well uh the case of continuous time active inference uh
uh i think it leads to uh

uh really interesting uh questions uh both in terms of uh philosophical questions and also uh
uh a more uh
uh practical modeling questions uh about uh what parameters uh needs to be accounted for and so on
uh uh and as i said i uh
uh believe uh it's a more uh
uh more interesting way of um

it's not uh
interesting but but at least more
uh involved way of um
uh doing active inference modeling
but uh one thing that uh
uh one of the uh
philosophical questions that um
uh
mao and i uh have explored
uh uh in our paper
uh is uh how
uh the processes
uh of uh i mean
ontological processes
uh can uh
philosophically described using
uh fvp uh
assertions
uh in terms of their
intra action with the environment
uh
in which they co-constitute themselves
and uh we don't necessarily
uh
distinguish between uh
between the internal and the external

so one obvious
example of this is
that generalized uh
synchrony example that we saw in this chapter
in which we don't
necessarily uh distinguish between
which of the birds act as
the agent and which one is the
uh environment uh or the vice

uh so these kind of
co-constitution of
the environment and the agent which
gives rise to
the partitioning of state space
through a markup blanket
is uh one of the
interesting
um philosophical
points uh that i think needs
to be uh elaborated
uh a bit more
uh using
some of the
recent advances
in uh philosophy such as
uh the two the tools
that's been developed in a new
materialism uh school
or some other philosophical
approach approaches but
uh yeah these kinds
of uh uh what exactly
gives size gives rise to
emergence what is the um ontological
status of emergent properties
and so on uh are some of the

um burning questions for many
philosophers today
and i believe active inference
and particularly continuous time active
inference provides
a clear
uh precise
mathematical formalism
uh
even uh if not to
answer these questions
but at least to explore
explore it in a more
rigorous and uh practical

and also
um
practical and attract
attractable way
so
uh this is
the area that i believe
philosophy and science
are beautifully intertwined
uh into a coherent view of
not only
uh the phenomena of interest
but even
about
the whole world
is

pretty cool
yeah a lot to say about
that topic

after completing chapter seven and eight you've seen the kind of two major branches or two major motifs of just one kind of modeling but these kind of models have so many different forms that that's why it's such a hands-on process to specify the generative model in chapter six and fit it with data in chapter nine those are all what's required and that's kind of the last mile of where these discussions about general motifs gets you but also playing with these pedagogical models can be really helpful because it will help you understand the basic patterns and relationships and start to see see different patterns in the graphical models and know from there what levels of technical processes can be kind of coarse grained over

you